



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester II Examinations 2022–2023

Course Instance Codes(s) 1SPS1, 3BA1, 3BMS2, 3BS9, 4BA16, 4BME1
Exam(s) 3rd Arts, 4th Arts, 3rd Science,

3rd BA Education, Masters.

Module Codes MA3491

Module **Fields and Applications**

Paper No 1

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Internal Examiner(s) Prof. Götz Pfeiffer
★ Dr Emil Sköldberg

Instructions:

Answer all FIVE questions

Duration 2 Hours

Number of Pages 3 (including this page)

Discipline Mathematics

Requirements:

Release in Exam Venue Yes ☒ No ☐

Release to Library Yes ☒ No ☐

Statistical/ Log Tables: State Examination Commission's *Formulae and Tables*

Handout None

Cambridge Tables None

Graph Paper None

Log Graph Paper None

Other Materials None

Graphic material in colour Yes ☐ No ☒

- Q1. (a) State the *sphere packing bound*, and determine the upper bound it gives on the number of code words in a 4-ary code with block length 8 and minimum distance 3. [6 marks]
- (b) State the *Singleton bound*, and determine the upper bound it gives on the number of code words in a ternary code with block length 8 and minimum distance 3. [6 marks]
- (c) Consider C , the binary block code with block length 8 which consists of all words with an equal number of 0s and 1s.
- (i) How many code words does C have? [4 marks]
- (ii) What is the minimum distance of C ? [4 marks]
- Q2. (a) For each of the following polynomials in $\mathbb{F}_5[x]$, determine if it is irreducible or not, and if it is reducible, factor it into irreducibles.
- (i) $x^3 + x^2 + 3$. [3 marks]
- (ii) $x^3 + x^2 + 4$. [3 marks]
- (b) (i) Show that $f(x) = x^3 + 2x + 1 \in \mathbb{F}_3[x]$ is irreducible, and hence let $\mathbb{F}_{27} = \mathbb{F}_3[x]/(f(x))$. [3 marks]
- (ii) Calculate the product $(x^2 + 1)(x^2 + x + 1)$ in $\mathbb{F}_{27}$. (Give the result as a polynomial of degree ≤ 2). [5 marks]
- (iii) Calculate the multiplicative inverse of x^2 in $\mathbb{F}_{27}$. (Give the result as a polynomial of degree ≤ 2). [6 marks]
- Q3. (a) Let C be the linear ternary code given by the generator matrix

$$G = \begin{pmatrix} 0 & 2 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

- (i) Write down the code words of C . [4 marks]
- (ii) Determine the minimum distance of C . [4 marks]
- (iii) What is the result of nearest-neighbour decoding of a received word 2102? [4 marks]
- (b) Consider the linear code C_2 over $\mathbb{F}_3$ with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 2 \end{pmatrix}.$$

- (i) Construct a check matrix for C_2 . [4 marks]
- (ii) Construct a syndrome table for C_2 and use it to decode the word 1111. [4 marks]

- Q4. (a) Let C be the linear $[5, 3]$ -code over $\mathbb{F}_3$ with check matrix

$$\begin{pmatrix} 1 & 0 & 2 & 0 & 1 \\ 2 & 1 & 0 & 2 & 1 \end{pmatrix}.$$

Determine the minimum distance of C . [6 marks]

- (b) (i) What is the definition of the Hamming code $Ham(r, q)$? [4 marks]
- (ii) Construct a parity check matrix for the Hamming code $Ham(2, 7)$. [4 marks]
- (iii) Explain how a single error can be corrected in $Ham(2, 7)$. Decode the word 11111111. [6 marks]

- Q5. (a) (i) Define the *weight enumerator* of a linear $[n, k]$ -code. [2 marks]
(ii) Compute the weight enumerator of the binary $[5, 2]$ -code given by the generator matrix

$$\begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

[6 marks]

- (b) Let C be the cyclic code of length 6 over the field $\mathbb{F}_5$ with generator polynomial $x^3 + 2x^2 + 2x + 1$.
(i) Construct the generator matrix for C . [4 marks]
(ii) Find the check polynomial for C and hence construct a check matrix for C . [8 marks]